

Some Attack Mitigations

Objectives:

At the end of this episode, I will be able to:

1. Describe some of the common attacks we face in the network
2. Provide examples of mitigation techniques to guard against these attacks

Additional resources used during the episode can be obtained using the download link on the overview episode.

Remember these common attack areas and mitigations:

Brute force password attacks

Lock the account

Delay between attempts

Password complexity rules

MFA

DNS

Harden servers

Log and review

Prevent use of open servers

Cisco Umbrella

Web

OWASP best practices

OS and browser versions

Cisco Umbrella

Cisco WSA

End user training

Command injection

Proper user input validation

IPS

SQL injections

Validate, constrain, and sanitize

IPS

Pre-build optimized code

XSS and CSRF

Cisco Umbrella

Cisco WSA

IPS

End user training

Web app best practices

Email

Cisco ESA

End user training

External Resources:

During this episode, you can reference the following external resources for supplementary tools and information:

You might know all of these more generic mitigation strategies already, but take a look at the NSA list of common attack mitigation techniques:

<https://www.nsa.gov/Portals/70/documents/what-we-do/cybersecurity/professional-resources/csi-nsas-top10-cybersecurity-mitigation-strategies.pdf>